FIGURE 09: MULTI-HIVE NETWORK SCALABILITY & ECONOMIC TOPOLOGY

100-Node Commercial Apiary: BLE Mesh Clusters + LoRa Backhaul Capacity Analysis & Zero Cellular Fee Advantage

APIARY LAYOUT (BLE MESH CLUSTERS + LORA BACKHAUL)

Commercial Apiary Yard: BLE Mesh Rows (2.4 GHz)

Channel Capacity & Duty Cycle Proof

[CALCULATED]

100 Hives organized in local 2.4 GHz BLE Mesh row clusters.

Designated cluster nodes uplink over Sub-GHz LoRa star link.

Packet airtime = 77.06 ms on channel (36-byte packed frame v2).

Aggregate Yard Duty Cycle = 1.82 s / 900 s = 0.202% (< 1.0% limit).

Dual-Radio Hybrid preserves low battery while extending yard range.

ECONOMIC & STRUCTURAL ARCHITECTURE COMPARISON

METRIC / CRITERION	COMMERCIAL CELLULAR	BEEVIL KNIEVEL
Sensor Node Hardware	Custom PCB (180–350)	**Modular WisBlock ($18.74)**
Monthly Cellular Fee / Hive	4.00–7.00 / month	**$0.00 (Zero SIM cards)**
100-Hive 3-Yr Telecom Cost	14, 400–25,200	**$0.00 Total**
Network Topology	Cellular Tower Dependency	**Dual: BLE Mesh + LoRa**
Brood RTD Precision	±0.5°C to ±1.0°C (Uncal)	**±0.1°C NIST (TMP117)**
Acoustic Feature Sampling	None or raw 2 kHz stream	**16 kHz I2S + CMSIS FFT**
Queenless Early Detection	Manual hive inspection	**Page's CUSUM (-0.02°C/hr)**
Cloud Dependency	Mandatory SaaS Portal	**100% Offline Local Gateway**

Scientific & Industry Distinction

[VALIDATED]

1 Academic Baseline: Ferrari et al. (2008) wired thermocouples (bulky, high-power).

2 Commercial Competitors: BroodMinder (BLE logger requiring manual yard walk) &

Arnia (Cellular hub with high initial cost and recurring monthly SaaS lock-in).

BEEVIL KNIEVEL bridges this gap with modular hardware, long-range LoRa & local edge AI.

[MEASURED] [VALIDATED] [CALCULATED] [SIMULATED] [DEMONSTRATED]